Semester Project Proposal Submission Form – FA'26 & onwards

DO NOT SUBMIT HANDWRITTEN FORM

FYP/CUI/CS-WAH/015

*** Project Title:**

(150 characters)

Comprehensive Data Structure Implementation & Analysis System (CDSIAS) with Learning Extension

*** Which real world problemshall be solved by this FYP?**

(800 characters)

Computer Science students often struggle to grasp the internal mechanics of data structures when relying solely on high-level libraries (like the STL). This creates a gap in understanding memory management, pointer manipulation, and algorithmic efficiency. CDSIAS bridges this gap by providing a "Digital Lab" environment where users can interact with manual implementations of linear and non-linear structures. It solves the lack of practical visualization and performance benchmarking tools available in standard academic settings.

Detail of Project Members:

- Akif Naveed FA24-BSE-129
- Arslan Shafiq FA24-BSE-119

Supervised By:

Dr .Wasif Nisar

Project Streams:

☐Web-based FYPs

☒Desktop Applications

☐Mobile Apps FYPs

☐Game-based FYPs

☐Hardware-based

*** Project Description:**

(3000 characters)

Note: Login/logout and authentication/authorization shall be the default functionalities of Web/Gaming/Mobile/Desktop FYPs.

(a) What is the overall working/summary of this FYP? (1000 characters)

The system operates as an interactive workbench. Upon secure login, users select a specific Data Structure module. They can perform real-time operations such as insertion, deletion, and searching. As these operations occur, the system tracks internal metrics (number of comparisons, pointer updates, and time taken). These statistics are stored in SQLite and displayed via the GUI to provide a quantitative understanding of algorithmic efficiency. The "Learning Extension" allows a Teacher to push assignments to Student accounts. Students complete these by implementing or interacting with the modules, and their performance data is automatically shared with the Teacher for evaluation.

(b) Write name and detail of each module in your FYP.

CDSIAS Modules: Features Linear Structures (Arrays/Lists) and Stack/Queues (Deques) for data handling. Tree (BST/AVL/Heaps) and Graph (BFS/DFS/Dijkstra) modules manage complex relations. Search/Sort and Hashing provide algorithmic analysis. A Performance engine uses SQLite to log metrics, while the Extension Module offers a Teacher-Student dashboard.

1: Which module shall be developed by student-1?

(500 characters)

Student 1 (Akif Naveed): Will develop the Tree Module (BST, AVL, Heap), Graph Module, and Hashing Module. He is responsible for implementing the back-end performance measurement logic and integrating these complex structures with the SQLite reporting engine.

2: Which module shall be developed by student-2?

(500 characters)

Student 2 (Arslan Shafiq): Will develop the Linear Structures Module, Stack & Queue Module, and the Searching/Sorting Module. He will also lead the GUI design in Qt and manage the database schema for user authentication and experiment logging.

3: Which module shall be developed by student-3?

(500 characters)

(d) Were similar FYPs already developed on the same topic in your department?

(●) Yes ( ) No

(d-1) Copy name(s) of latest one or two similar FYPs from RMS student console and paste below. (150 characters)

(d-2) Mention below the three (3) new, but main, functionalities you are adding to this FYP. (600 characters)

1.

2.

3.

Development Environment:

- **Tools** (e.g.Dot Net platform, Android Studio,
- Xcode, Swift, Ionic, Xamarin, PhpStorm
- /Php Laravel, WordPress, Maya, Unity 3D,
- Photoshop, MATLAB, ns-2, Python,
- Java EE, Java ME, NetBeans, Java Script,
- Node.js, Angular.js, JSON, OpenCV)
- **DBMS** (e.g. SQL Server, MySQL, SQLite,
- Oracle, Teradata)
- **Platform** (e.g. Windows, Linux

- **Tool(s):** C++ (Qt Framework), Qt Creator / VS Code, GitHub
- **DBMS:** SQLite
- **Platform:** Windows

Evaluation Criteria for 7th Semester:

(Week 14 - 16)

1. SRDS (Functional & Non-functional requirements, Use case diagram, Sequence diagram, Class diagram, Entity relationship diagram / Detailed hardware configuration)

2. Implementation of ONE major use case (>=30% FYP work), which does not includelogin/logout.

3. Interface with complete functionality of major use case / In case of hardware, provideconfiguration of major use case functionality.

Evaluation Criteria for Internal 8th Viva:

(Week 12 & 13)

1. Implementation of all use cases (>=90% FYP work)

2. Project in running and working form as per Use case, Class and Sequence diagrams asmentioned in SRDS

3. Initial FYP report

4. Deployment
   - Web: Your website must be online on any free/paid hosting service
   - Mobile App: Your app must be in APK/iOS form so that it can be installed
   - Gamimg: Your game must be in executable form so that it can be installed
   - Desktop Application: Your application must be in executable form so that it can beinstalled
   - Hardware: Your final product must be in proper casing and should give look and feelof sellable item